

Student Performance Prediction using Machine Learning

Objective

The objective of this project is to build and evaluate machine learning models for predicting student mathematics scores based on demographic and academic performance factors. The project applies data preprocessing, feature selection, model training, and evaluation techniques to identify the most effective prediction model.

Dataset Description

The Student Performance Dataset contains information about students and their academic performance. The dataset includes the following attributes:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch Type
- Test Preparation Course
- Reading Score
- Writing Score
- Math Score (Target Variable)

Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the dataset using Pandas.
2. Checked for missing values and data consistency.
3. Converted categorical variables into numerical values using Label Encoding.
4. Prepared the dataset for machine learning model training.

Feature Selection

The target variable selected for prediction was:

Math Score

The remaining columns were used as input features to train the machine learning models.

Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

This split ensures that the models are trained on one portion of the dataset and evaluated on unseen data.

Machine Learning Models

1. Linear Regression

Linear Regression was applied to establish a relationship between input features and student math scores.

2. Decision Tree Regressor

Decision Tree Regressor was used to capture non-linear relationships within the dataset and improve prediction performance.

3. Random Forest Regressor

Random Forest Regressor combined multiple decision trees to improve prediction accuracy and reduce overfitting.

Model Evaluation

The models were evaluated using:

Root Mean Squared Error (RMSE)

RMSE measures the average prediction error. Lower RMSE values indicate better model performance.

R² Score

R² Score measures how well the model explains the variation in the target variable. Higher values indicate better performance.

Results

The performance of all models was compared using RMSE and R² Score metrics.

The model with:

- Lowest RMSE
- Highest R² Score

was considered the best-performing model.

In most cases, the Random Forest Regressor achieved better predictive performance than Linear Regression and Decision Tree models.

Key Insights

1. Reading Score and Writing Score have a strong influence on Math Score.
2. Students with better overall academic performance tend to achieve higher mathematics scores.
3. Machine Learning models can successfully predict student outcomes based on historical data.
4. Random Forest generally provides higher prediction accuracy compared to individual regression models.

Conclusion

This project successfully developed and evaluated machine learning models for student performance prediction. The results demonstrate that machine learning techniques can be effectively used to analyze educational data and predict student outcomes. Such predictive systems can assist educational

institutions in identifying students who may require additional academic support and help improve overall learning outcomes.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Google Colab

Screenshots:

